SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 6

Name: **杨涵**
Student ID: 2025531064

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(a) For $X \sim \text{Pois}(\lambda)$ and any function g, derive the relationship between $E[Xg(X)]$ and $E[g(X+1)]$. This is called the Stein-Chen identity for the Poisson.

(b) Find the moment $E\left(X^4\right)$ for $X \sim \text{Pois}(\lambda)$ by using the identity from (a) with the fact that X has mean λ and variance λ.

Solution

(a) The $k = 0$ term in the expectation is zero. Starting the sum at $k = 1$ and then shifting the index gives

$$\begin{aligned}
\mathrm{E}[Xg(X)] &= \sum_{k=1}^{\infty} kg(k)e^{-\lambda}\frac{\lambda^k}{k!} \\
&= \lambda\sum_{k=1}^{\infty} g(k)e^{-\lambda}\frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda\sum_{k=0}^{\infty} g(k+1)e^{-\lambda}\frac{\lambda^k}{k!} \\
&= \lambda\mathrm{E}[g(X+1)].
\end{aligned}$$

(b) Since $\mathrm{E}[X] = \lambda$ and $\mathrm{Var}(X) = \lambda$,

$$\mathrm{E}[X^2] = \mathrm{Var}(X) + (\mathrm{E}[X])^2 = \lambda + \lambda^2.$$

Applying the identity from (a) with $g(x) = x^2$ gives

$$\begin{aligned}
\mathrm{E}[X^3] &= \lambda\mathrm{E}[(X+1)^2] \\
&= \lambda\big(\mathrm{E}[X^2] + 2\mathrm{E}[X] + 1\big) \\
&= \lambda^3 + 3\lambda^2 + \lambda.
\end{aligned}$$

Now take $g(x) = x^3$. Then

$$\begin{aligned}
\mathrm{E}[X^4] &= \lambda\mathrm{E}[(X+1)^3] \\
&= \lambda\big(\mathrm{E}[X^3] + 3\mathrm{E}[X^2] + 3\mathrm{E}[X] + 1\big) \\
&= \lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda.
\end{aligned}$$

Problem 2

Alice flips a fair coin n times and Bob flips another fair coin $n+1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n+1, \frac{1}{2})$.
(a) Let $V = \min(X, Y)$ be the smaller of X and Y, and let $W = \max(X, Y)$ be the larger of X and Y. (If $X = Y$, then $V = W = X = Y$.) Find $E(V) + E(W)$.
(b) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.
(c) Compute $P(X < Y)$. Hint: Use (b) and the fact that X and Y are integer-valued.

Solution

(a) Since $V + W = X + Y$, linearity of expectation gives

$$\mathrm{E}[V] + \mathrm{E}[W] = \mathrm{E}[X] + \mathrm{E}[Y] = \frac{n}{2} + \frac{n+1}{2} = n + \frac{1}{2}.$$

(b) Swapping heads and tails in every flip produces a sequence with the same probability. Under this swap, the head counts (X, Y) become the tail counts $(n - X, n + 1 - Y)$. Thus the two pairs have the same joint distribution, and

$$P(X < Y) = P(n - X < n + 1 - Y).$$

(c) Rearranging $n - X < n + 1 - Y$ gives $Y - X < 1$. Since X and Y are integer-valued, this is equivalent to $Y - X \leq 0$, or $Y \leq X$. Hence $P(X < Y) = P(Y \leq X)$ by (b). The two events are complementary, so

$$P(X < Y) = \frac{1}{2}.$$

Problem 3

Consider two probability distributions $P = (0.75, 0.25)$ and $Q = (0.5, 0.5)$ over a binary random variable $X \in \{0, 1\}$. All logarithms are base 2.

(a) **Basic Calculations (5 points)**

Compute the total variation distance and KL divergence:

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|,$$
$$D_{\mathrm{KL}}(P\|Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}.$$

Is KL divergence symmetric? Justify your answer.

(b) **Connection to Information Theory (5 points)**

(i) Compute the entropy $H(P)$ and cross-entropy $H(P, Q)$, which are defined as

$$H(P) = -\sum_x P(x) \log_2 P(x),$$
$$H(P, Q) = -\sum_x P(x) \log_2 Q(x).$$

(ii) Show that

$$H(P, Q) - H(P) = D_{\mathrm{KL}}(P\|Q).$$

Briefly explain the significance of this identity in lossless data compression.

(c) **Connection to Machine Learning (5 points)**

Suppose P represents the true label distribution in a binary classification task. Two models predict $Q = (0.5, 0.5)$ and $R = (0.95, 0.05)$, respectively.

Compute $D_{\mathrm{KL}}(P\|Q)$ and $D_{\mathrm{KL}}(P\|R)$. Which model has a smaller expected cross-entropy loss? Explain your answer.

Solution

(a) The total variation distance is

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)| = \frac{1}{2}\big(|0.75 - 0.5| + |0.25 - 0.5|\big) = 0.25.$$

The KL divergences in the two directions are

$$D_{\mathrm{KL}}(P\|Q) = 0.75 \log_2 \frac{0.75}{0.5} + 0.25 \log_2 \frac{0.25}{0.5} \approx 0.1887 \text{ bits},$$
$$D_{\mathrm{KL}}(Q\|P) = 0.5 \log_2 \frac{0.5}{0.75} + 0.5 \log_2 \frac{0.5}{0.25} \approx 0.2075 \text{ bits}.$$

These values differ, so KL divergence is not symmetric.

(b) (i) The entropy and cross-entropy are

$$H(P) = -0.75 \log_2 0.75 - 0.25 \log_2 0.25 \approx 0.8113 \text{ bits},$$
$$H(P, Q) = -0.75 \log_2 0.5 - 0.25 \log_2 0.5 = 1 \text{ bit}.$$

(ii) Subtracting the entropy from the cross-entropy gives

$$\begin{aligned} H(P, Q) - H(P) &= -\sum_x P(x) \log_2 Q(x) + \sum_x P(x) \log_2 P(x) \\ &= \sum_x P(x) \log_2 \frac{P(x)}{Q(x)} = D_{\mathrm{KL}}(P \| Q). \end{aligned}$$

In lossless data compression, $D_{\mathrm{KL}}(P\|Q)$ is the extra number of bits per symbol, on average, for ideal coding based on Q when the true source follows P, compared with ideal coding based on P.

(c) From (a), $D_{\mathrm{KL}}(P\|Q) \approx 0.1887$ bits. For R,

$$D_{\mathrm{KL}}(P\|R) = 0.75 \log_2 \frac{0.75}{0.95} + 0.25 \log_2 \frac{0.25}{0.05} \approx 0.3247 \text{ bits}.$$

For a model distribution M, the expected cross-entropy loss is $H(P, M) = H(P) + D_{\mathrm{KL}}(P\|M)$. Since $H(P, Q) = 1$ bit and $H(P, R) \approx 1.136$ bits, the model predicting Q has the smaller expected cross-entropy loss.